and extending to the centre is called **ground tissue**. It is not differentiated into endodermis, cortex, pericycle and pith.

4. **Vascular Bundle:** Vascular bundles are skull shaped and scattered in the ground tissue. Vascular bundles are conjoint, collateral, endarch and closed. Each vascular bundle is surrounded by few layer of sclerenchyma cells called **bundle sheath**.
   - **(a) Xylem:** It consists of metaxylem and protoxylem. Xylem vessels are arranged in V or Y shape. In mature vascular bundle, the lower most protoxylem disintegrates and form a cavity. This is called **protoxylem lacuna.**
   - **(b) Phloem:** It consists of sieve tube elements and companion cells. Phloem parenchyma, and phloem fibers are absent.
5. **Pith:** Pith is not differentiated in monocot stems.

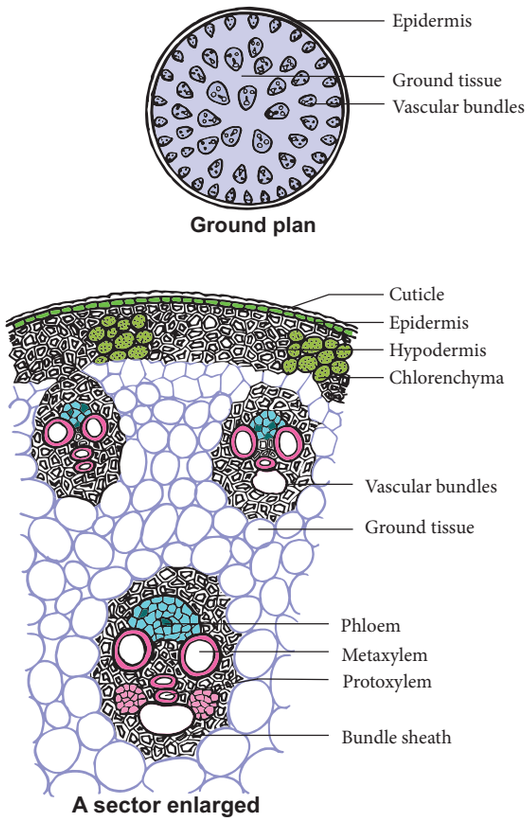


Figure 12.5 Transverse section of Monocot stem

Table 12.3 Differences between Dicot Stem Ex.Sunflower and Monocot Stem Ex.Maize

S. No.	Tissues	Dicot Stem	Monocot Stem
1	Hypodermis	Collenchymatous	Sclerenchymatous
2	Ground tissue	Differentiated into cortex, endodermis, pericycle and pith	Undifferentiated
3	Vascular bundles	(i) Less in number (ii) Uniform in size (iii) Arranged in a ring (iv) Open (Cambium present) (v) Bundle sheath absent	(i) Numerous (ii) Smaller near periphery, bigger in the centre (iii) Scattered (iv) Closed (Cambium absent) (v) Bundle sheath present
4	Secondary growth	Present	Mostly absent
5	Pith	Present	Absent
6	Medullary rays	Present	Absent

12.7 Internal Structure of Dicot Leaf (Dorsiventral Leaf) Ex. Mango Leaf

The transverse section of leaf shows the following structures.

- **(i) Upper epidermis:** This is the outermost layer made of single layered parenchymatous cells without intercellular spaces. The outer wall of the cells are cuticularized. Stomata are less in number.

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